3 **(a)** By conservation of energy, loss in epe of spring = Gain in ke of block A

M1

$$\frac{1}{2}kx^2 - 0 = \frac{1}{2}(m_A)u^2 - 0$$ A1

$$k(0.020)^2 = (1.5)(0.50)^2$$

$$k = 938 \text{ N m}^{-1}$$

(b) By conservation of momentum: $m_A u = m_A v_1 + m_B v_2$ M1

$1.5(0.50) = (1.5)v_1 + (0.50)v_2 \Rightarrow 0.75 = 1.5v_1 + 0.50v_2 \;--(1)$

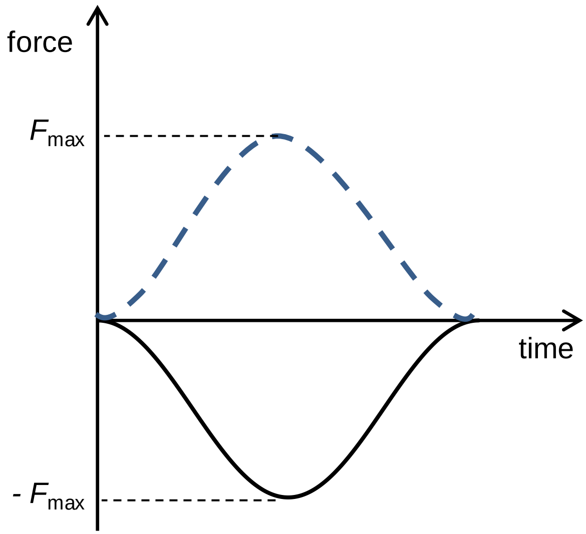
Relative speed of approach = relative speed of separation M1

$0.5 = v_2 - v_1 \Rightarrow v_1 = v_2 - 0.5 \;--(2)$

solve, $v_2 = 0.75 \text{ m s}^{-1}$ A1

[The math is much more tedious if students chose to work with conservation of KE equation. For head-on elastic collision, students can use 'Relative speed of approach = relative speed of separation']

(c) **(i)**



B1

(ii) Area under the force time graph is change in momentum of a body. B1

Area under the graphs of A and B are equal in magnitude but opposite in direction. This implies gain in momentum of one block equals loss in momentum of the other. B1

(d) B continues to move after colliding with A again and hence has kinetic energy. A has less kinetic energy than at the start as part of it has been transferred to B. Thus energy transferred to spring is lesser and the new compression of spring will be less than 2.0 cm. M1 A1